



BUSINESS BLUEPRINT

AOP AUTOMATION FOR ELASTOMER JMD HIIR PLANT

Honeywell

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Business Blueprint

AOP Automation for Elastomer JMD HIIIR Plant

Prepared for:

Reliance Industries Ltd

Submitted by:

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Revision History

Rev.	Date	Originated By	Description of status and changes
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1. Introduction

The purpose of this document is to capture the detailed specifications for Digitalized Annual Operating Plan (AOP) for HIIR plant under Elastomer vertical.

This document is based on the Technical Proposal submitted by Honeywell to Reliance. It also refers to data collected during the workshop conducted at RCP with various Elastomer plant users across RIL sites.

1.1 Executive Summary

Reliance Industries Limited is India's leading private sector organization operating in various business verticals. The manufacturing sector is one such business vertical. RIL owns and operates multiple plants across all sites.

Defining and finalizing the Annual Operating Plan is a critical annual business process carried out by different functional teams. At present this activity is carried out manually.

A Rev-0 budget is usually completed in Nov / Dec for the succeeding Financial Year (FY) and reviewed. The comments are addressed with updated information from the Business as the next revised budget by the end of Jan. Further revisions (if any) are incorporated in Feb / March to reflect updated business input as applicable.

The entries from the final rev of the budget are finalized in BPC before the beginning of the succeeding FY.

RIL aims to automate the AOP preparation process using a Data-Driven Model. This model leverages the last five years of AOP data from MIIS, PAS, SAP, IP21, LIMS and incorporates plant-specific business rules (such as plant availability, running hours, feedstock availability, mechanical/maintenance slowdown activity, etc) to derive AOP figures in the expected formats.

As part of digitalization, Honeywell will support building a platform that will automate the Annual Operations Planning process for PET technology.

1.2 Reference Documents

Document ID	Revision	Title	Document
O-836243B24R01	R01	AOP Automation for O2C Proposal	 O-836243B24R01 RIL AOP Automation f
Elastomer JMD IIR_HIIR DCT	-	Elastomer DCT IIR_HIIR	 Elastomer_IIR_HIIR_D CT

PAS Consumption & Production Tags	-	Elastomer DTA HIIR PAS daily consumption & production	 HIIR PAS Consumption & Prodi
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Table 1 Reference Documents

2. AOP Design Basis

The below table captures different functions for the scope of AOP digitization design –

Sr. No.	Functions	Covered in Document
1.	Health, Safety and Environment (HSE)	For later release post review and discussion
2.	Process/Operations	Covered
3.	Reliability and Maintenance	For later release post review and discussion

Table 2 AOP Functional Scope

Function-wise detailed design basis is captured in the below sections.

2.1 Process and Operations Function

In this section, the budgeted Production and Norms for a plant are derived, using a standard approach as briefed in the sections below for the following criteria

- Budget Production/Production AOP
- Norms/Consumption AOP

Steps to generate the Production and Consumption AOP are captured in the below flow diagram.

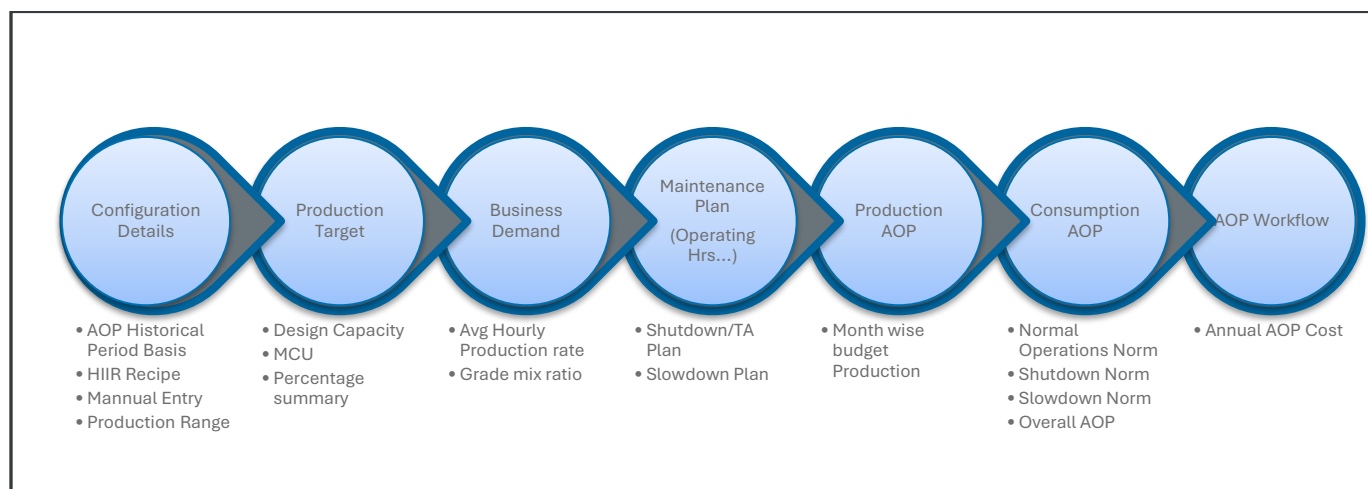


Figure 1 Digitized AOP Flow Diagram

2.1.1 Budget Production / Production AOP

Below are the steps captured to derive the budgeted AOP production values.

2.1.1.1 Production & Norms Basis

The configuration screen serves as the foundation for the digital AOP process, allowing users to input essential data for production and consumption AOP calculations.

Plant-specific data are required to calculate budget production, consumption norms, and reports generation.

By default, the values entered for current year will automatically be carried forward to the next year.

Additional features and tabs are provided as follows:

- **AOP historical period basis** added where Start and End date is provided to select historical period to calculate MCU and Norms data for selected period by default 03 or 05 years data will be loaded. AOP design basis will reflect the selected period as AOP basis, users can add additional basis points for summary.
- **Configuration tab** is provided for users to enter grade wise month wise packaging material & quality data manually
- **Constant tab** is provided for users to define HIR grade wise recipe
- **Production Range tab** is provided for users to define the production range for MCU calculation & utility norms calculation
- **Import and export** functions are available on each tab for bulk update

The screenshot below highlights the details captured in the digitized AOP platform

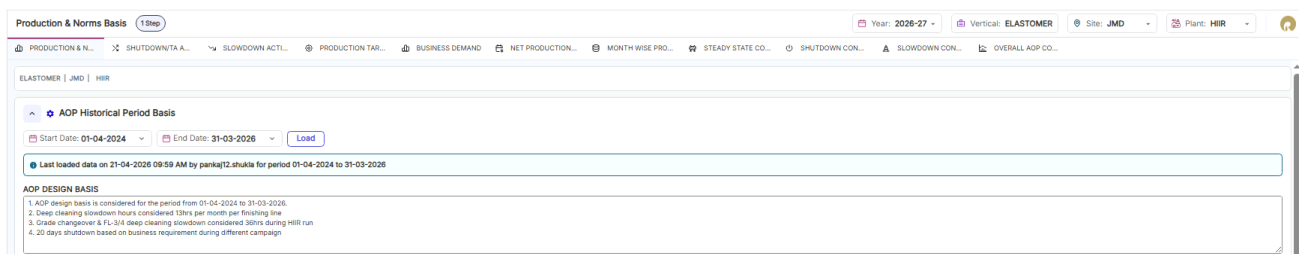


Figure 2 AOP Historical Period Basis

Configuration Constant ProductionRange														
Configuration 57 Rows														
Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
CIIR - Packaging Material														
BAGS,HDPE WOVEN FOR CIIR 1139 INNER ...	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
CAP FOR MB-5 CRATE	EA	40.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
CRATE LINER,SIZE:180"X55", BLUE	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
CRATE LINER,SIZE:200"X42", BLUE	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
CRATE, PLYWOOD, BALE PACKING	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
CRATE, PLYWOOD, BALE PACKING	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
EVA FILM, EMBOSSED, FOR BALE PACKAG...	KG	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
FILMS,EVA FOR CI139-HIIR WITH NEEDLING	KG	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
FILMS,EVA FOR CIIR-1139-HIIR BALE PACK...	KG	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
LABEL,CRATE,SIZE:150X100 MM- CIIR	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
LABLE,BARCODE,SIZE:100X40 MM	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...
RIBBON-A, DOMINO BLACK,SIZE:110 MM	EA	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	25.0000	Manual Entry...

Figure 3 Configuration

Configuration Constant ProductionRange				
Constant 17 Rows				
Particulars	UOM	Value	Remark	
Raw Material				
IIR IMPRAMER R 1675	MT	1.0000	Default Value	
HIIR Recipe - CI139				
IRGANOX 1010 / ANOX 20	KG/MT	0.7000	Manual Entry	
CHEM CALC STEARATE DISPERSION	KG/MT	20.0000	Manual Entry	
CHEM CALCIUM CHLORIDE (CACL2)	KG/MT	2.5000	Manual Entry	
CHEM CHLORINE	KG/MT	29.0000	Manual Entry	
EPOXIDIZED SOYABEAN OIL	KG/MT	5.0000	Manual Entry	
CHEM NONYLPHENOL-(RING-UL-14C) 4-ET	KG/MT	0.0300	Manual Entry	
CHEM SODIUM SULPHITE,PN:MIS 190X	KG/MT	5.0000	Manual Entry	
HIIR Recipe - B2232/B2247				
IRGANOX 1010 / ANOX 20	KG/MT	0.7000	Manual Entry	
CHEM CALC STEARATE DISPERSION	KG/MT	40.0000	Manual Entry	
CHEM CALCIUM CHLORIDE (CACL2)	KG/MT	2.5000	Manual Entry	

Figure 4 Constant

Configuration Constant ProductionRange					
Production Range 5 Rows					
Particulars	UOM	Min	Max	Remark	
B2232 - Production Range for MCU					
CIIR - Production Range For Utility -1	TPD	175	190	Manual Entry	
CIIR - Production Range For Utility -2	TPD	160	180	Manual Entry	
BIIR - Production Range For Utility -1	TPD	100	120	Manual Entry	
BIIR - Production Range For Utility -2	TPD	145	175	Manual Entry	
Production Range (Limit) 3 Rows					
Particulars	UOM	Limit	Value	Remark	
CI139 - Production Range for MCU	TPD	>=	180	Manual Entry	
B2247 - Production Range for MCU	TPD	>=	95	Manual Entry	
B2247 - Production Range For Utility	TPD	>=	80	Manual Entry	

Figure 5 Production Range

2.1.1.2 Production Target

HIIR Max Achieved Capacity (MCU) is derived using last 02 years of historical data from PAS system. Detailed calculation steps are mentioned as follows:

1. Select AOP historical period basis and load the data for past 2 years
2. Historical data of following tags are loaded from PAS

B2232 Production, kg	C1139 Production, kg
HIIR_OUT001	HIIR_OUT006
HIIR_OUT002	HIIR_OUT007
HIIR_OUT003	HIIR_OUT008
HIIR_OUT004	HIIR_OUT009
HIIR_OUT005	HIIR_OUT010

3. Calculate day wise B2232 & C1139 production using formula as below

B2232	$SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+HIIR_OUT005)/1000$
C1139	$SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+HIIR_OUT010)/1000$
	All units in MT

Grade C1139 MCU Calculation:

1. Filter C1139 production days where production $\geq$ 180 TPD
2. Prepare multiple clusters for a minimum of 5 consecutive days per cluster
3. Take moving average of each cluster
4. Filter top 5 moving average values of formed clusters
5. Calculate normal average of top 5 moving average values of formed clusters
6. **Grade C1139 MCU** = normal average of top 5 moving average values of formed clusters

Grade B2232 MCU Calculation:

1. Filter B2232 production days where production is between 175 - 190 TPD
2. Calculate normal average of filtered production datapoints
3. **Grade B2232 MCU** = normal average of filtered production datapoints

Grade B2247 MCU Calculation:

1. Fetch day wise (posting date) B2247 production quantity from SAP [KG]
2. Convert day wise production quantity from KG to MT
3. Filter B2247 production days where production $\geq$ 95 TPD
4. Prepare multiple clusters for a minimum of 5 consecutive days per cluster
5. Take moving average of each cluster
6. Filter top 5 moving average values of formed clusters
7. Calculate normal average of top 5 moving average values of formed clusters
8. **Grade B2247 MCU** = normal average of top 5 moving average values of formed clusters

HIIR MCU Calculation:

HIIR MCU = Normal average of C1139 MCU and B2232 MCU

Note: AOP historical period basis added in configuration screen where user can change start date and end date to change the AOP design basis. Accordingly, MCU and Norms historical data will be considered for calculation.

The following logic has been formulated to derive MC values.



Figure 6 Digitized AOP Steps to derive MC value for Grade C1139

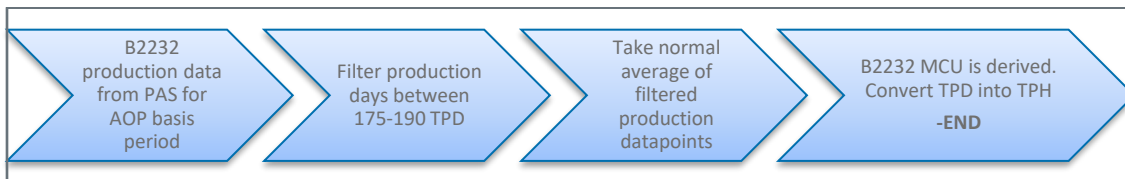


Figure 7 Digitized AOP Steps to derive MC value for Grade B2232



Figure 8 Digitized AOP Steps to derive MC value for Grade B2247

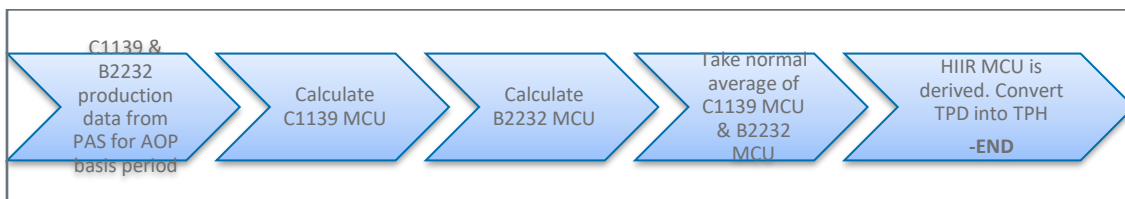
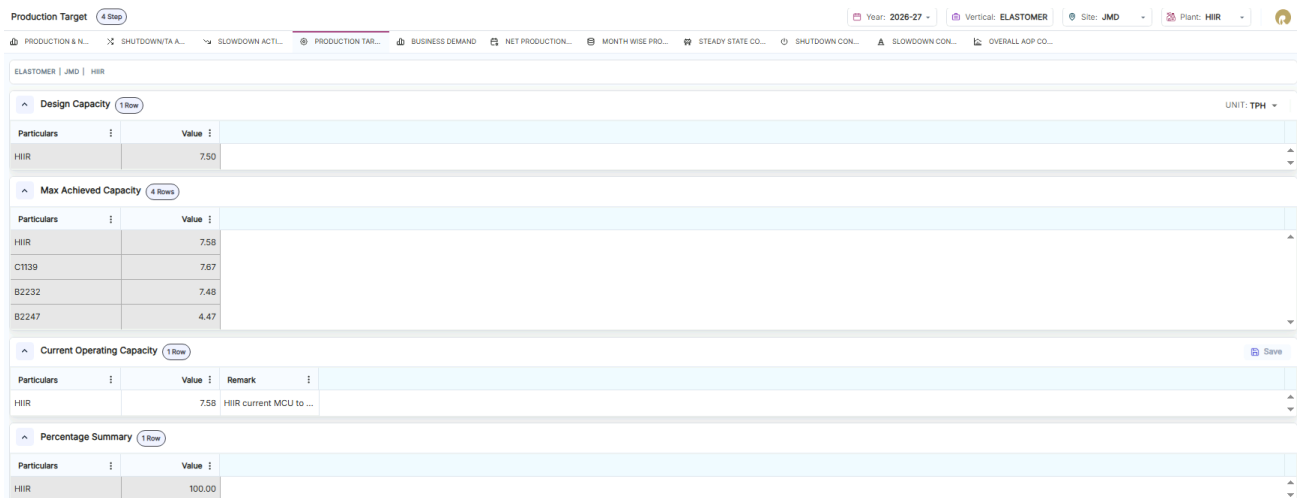


Figure 9 Digitized AOP Steps to derive MC value for HIIR

If the user needs to modify the HIIR max achieved capacity, this can be done by providing a specific reason and then modified MCU shall be the proposed operating capacity (MCU) for Average Hourly Production Rate calculation.

The AOP digitization portal captures percentage summary of HIIR proposed operating capacity, comparing value against the highest MC value recorded over the past 12 months.

This information is captured in the AOP digitization portal as seen in the screenshot below.



Production Target		
ELASTOMER JMD HIIR		
Design Capacity (1 Row)		
Particulars	Value	
HIIR	7.50	
Max Achieved Capacity (4 Rows)		
Particulars	Value	
HIIR	7.58	
CT139	7.67	
B2232	7.48	
B2247	4.47	
Current Operating Capacity (1 Row)		
Particulars	Value	Remark
HIIR	7.58	HIIR current MCU to ...
Percentage Summary (1 Row)		
Particulars	Value	
HIIR	100.00	

Figure 10 Digitized AOP screen for Production Target

Average Hourly Production Rate:

HIIR annual budget production is calculated based on budgeted average hourly production rate and plant operating hours. The calculation of average hourly production rate and plant operating hours are derived as follows.

- 1- [Plant Operating Hours] = [Available Hours in Year] – [Total Plant Shutdown Hours]
- 2- [Effective Operating Hours] = [Plant Operating Hours] – [Total Slowdown Hours]
- 3- Achievable Hourly Production rate for each slowdown activity is defined in AOP
- 4- Slowdown Hours for each slowdown activity is defined in AOP
- 5- Calculate sum product of slowdown hours and achievable hourly production rate during slowdown
- 6- Calculate the product of HIIR MCU (TPH) and effective operating hours

Budget Average Hourly Production Rate (TPH)

$$= \frac{(\text{Sum Product of slowdown hours \& Achievable hourly production rate})}{(\text{Product of HIIR MCU \& Effective operating hours})}$$

Average Hourly Production Rate (1 Row)		UNIT: TPH
Particulars	Value	
HIR	6.91	

Figure 11 Budget Average Hourly Production Rate

2.1.1.3 Business Demand

The grade mix demand will be provided by the Business team and added in AOP portal manually. In the AOP digitization application, annual grade business demand for each individual grade will be recorded.

HIR Product Grades:

- C1139
- B2232
- B2247

An example of the same is mentioned in the table below.

Business Demand5 Step

Year: 2026-27

Vertical: ELASTOMER

Site: JMD

Plant: HIR

PRODUCTION & N...

SHUTDOWN/TA A...

SLOWDOWN ACTL...

PRODUCTION TAR...

BUSINESS DEMAND

NET PRODUCTION...

MONTH WISE PRO...

STEADY STATE CO...

SHUTDOWN CON...

SLOWDOWN CON...

OVERALL AOP CO...

ELASTOMER | JMD | HIR

^

Average Hourly Production Rate

1 Row

UNIT: TPH

Particulars	Value
HIR	6.91

^

Business Demand

3 Rows

Save

Particulars	Value	Remark
Production		
C1139	60	Manual Entry
B2232	27	Manual Entry
B2247	13	Manual Entry
Total	100	0

Figure 12 Grade-wise business demand

2.1.1.4 Shutdown / Slowdown Activity

The operations team will provide the shutdown plan to calculate annual plant operating hours.

User shall select the shutdown month from dropdown and shutdown hours for each shutdown activity in digitized AOP shutdown / TA activities screen.

The operations team shall provide the slowdown plan. Slowdown activities shall be plant wise and NOT grade wise. Grade changeover and deep cleaning kind of activities shall be planned as slowdown activities.

Total Plant Operating Hours = Available hours in a year – Total Plant Shutdown hours)

Total Plant Slowdown Hours = Sum of slowdown hours of each slowdown activity

$$\text{Total Effective Operating Hours} = \text{Total Plant Operating Hours} - \text{Total Plant Slowdown Hours}$$

The screenshot below depicts the shutdown activities planned, as will be captured in the AOP digitized platform.

ShutDown/TA Activities (2 Step)

Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: HIR

ShutDown/TA Activities (4 Rows)

ShutDown Desc	Month	Duration (hrs)	Remarks	Action
Routine Shutdown-1	June	720:00	routine shutdown	
Routine Shutdown-2	July	720:00	routine shutdown	
Routine shutdown-4	October	360:00	routine shutdown	
Routine shutdown-5	November	720:00	routine shutdown	

Figure 13 Digitized AOP screen for shutdown activities

The screen below depicts slowdown activities planned as will be captured in the digitized platform.

Slowdown Activities (3 Step)

Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: HIR

Slowdown Activities (3 Rows)

Slowdown Desc	Duration (hrs)	Rate (TPH)	Remarks	Action
Deep Cleaning	130:00	3	Deep Cleaning	
FL-3/4 Grade Changeover	324:00	0	FL-3/4 Grade Chang...	
B2247 Slowdown	300:00	2	B2247 Slowdown	

Figure 14 Digitized AOP screen for slowdown activities

The screenshot below captures an example of the number of operating hours, shutdown hours, and effective slowdown hours available annually.

Net Production Hours (6 Step)

Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: HIR

Net Production Hours (5 Rows)

Description	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Total
Total Available Hours for the Month	720.00	744.00	720.00	744.00	744.00	720.00	744.00	720.00	744.00	744.00	672.00	744.00	8,760.00
Shutdown Hours	-	-	720.00	720.00	-	-	360.00	720.00	-	-	-	-	2,520.00
Operating Hours	720.00	744.00	-	24.00	744.00	720.00	384.00	-	744.00	744.00	672.00	744.00	6,240.00
Slowdown Hours	-	-	-	-	-	-	-	-	-	-	-	-	754.00
Effective Operating Hours	-	-	-	-	-	-	-	-	-	-	-	-	5,486.00

Figure 15 Net Production Hours

Note: RIL to validate slowdown and shut down hours calculated in System and raise any exception observed.

2.1.1.5 Calendar Year Based Budget Production

Using the budget average hourly production rate, and total plant operating hours, HIIR annual budget production quantity will be derived.

Grade wise annual budget production shall be derived using HIIR annual budget production and grade wise business demand.

HIIR Annual Production

$$= (\text{Budget average hourly production rate}) * (\text{Total plant operating hours})$$

$$\text{Grade C1139 Annual Production} = (\text{HIIR annual production}) * (\text{C1139 business demand in \%})$$

$$\text{Grade B2232 Annual Production} = (\text{HIIR annual production}) * (\text{B2232 business demand in \%})$$

$$\text{Grade B2247 Annual Production} = (\text{HIIR annual production}) * (\text{B2247 business demand in \%})$$

Grade R1675 Annual Production

$$= (\text{Total IIR annual production}) - (\text{C1139 Annual Production}) \\ - (\text{B2232 Annual Production}) - (\text{B2247 Annual Production})$$

Note:

1. HIIR annual production & Grade wise annual production are calculated and displayed in HIIR plant digital AOP, however they are displayed in IIR plant digital AOP also.
2. Grade wise month wise production quantity shall be entered manually in IIR plant digital AOP. However, they are also displayed as a read-only in HIIR plant digital AOP

The screenshot below captures the calculated annual budgeted production and grade wise month wise production quantity.

Month wise Production plan

7 Step

Year: 2026-27

Vertical: ELASTOMER

Site: JMD

Plant: HIIR

PRODUCTION & N...

SHUTDOWNTA A...

SLOWDOWN ACTL...

PRODUCTION TAR...

BUSINESS DEMAND

NET PRODUCTION...

MONTH WISE PRO...

STEADY STATE CO...

SHUTDOWN CON...

SLOWDOWN CON...

OVERALL AOP CO...

ELASTOMER

JMD

HIIR

HIIR Annual production

5 Rows

UNIT: MT

Export

Product	Values
HIIR Annual Production	
HIIR	42557.65
Grade wise Annual Production	
C1139	25534.59
B2232	11490.56
B2247	5532.49
R1675	75842.58

Month wise Production plan

5 Rows

UNIT: MT

Export

Products	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Total	Remark
C1139	600.00	700.00	800.00	0.00	3636.98	26.61	1100.00	3621.00	11000.00	3200.00	500.00	350.00	25534.59	Updated
B2247	50.00	4697.97	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	50.00	734.52	5532.49	Updated
B2232	0.00	9000.00	861.11	0.00	0.00	0.00	0.00	0.00	0.00	1629.45	0.00	0.00	11490.56	Updated
R1675	0.00	0.00	64.52	5000.00	70000.00	0.00	0.00	777.83	0.00	0.23	0.00	0.00	75842.58	Updated.
Total	650.00	14397.97	1725.63	5000.00	73636.98	26.61	1100.00	4398.83	11000.00	4829.68	550.00	1084.52	118400.22	Add

Figure 16 Budgeted Production Digitized AOP Screen

2.1.2 Norms

Below are the steps captured to derive the Norms value.

2.1.2.1 State-State Operations

Grade wise steady state norms are derived for following HIIR grades. However, overall AOP consumption norms are derived as a combined grade i.e. 'HIIR Combined'. The detailed logic is explained below in subsequent sections.

1) CIIR - C1139 2) BIIR - B2232 3) BIIR - B2247

- For Utilities, steady state consumption quantity is derived for each utility particulars
- For Cat-Chem, steady state consumption norms are derived for each Cat-Chem particular
- For Packaging materials, steady state consumption norms shall be manual entry
- For Quality materials, steady state consumption norms shall be manual entry

Raw Material (Source: Constant Value):

1. IIR IMPRAMER R1675 production is the raw material for all the HIIR grade wise production
2. There shall be a constant value of IIR R1675 raw material i.e. 1MT for all the HIIR grade wise production

Catalyst & Chemicals (Source: SAP):

For certain catalysts & chemicals as follows, base norms are provided as grade wise HIIR recipe manual entry. The steady state norms of such chemicals are derived from SAP historical data. However, derived norm, which is between defined range of base norm, is taken as steady state consumption norms.

HIIR Recipe – C1139 (Manual Entry)

HIIR Recipe - C1139	UoM	UOM/MT
IRGANOX 1010 / ANOX 20	KG	0.7
CHEM CALC STEARATE DISPERSION	KG	20
CHEM CALCIUM CHLORIDE (CACL2)	KG	2.5
CHEM CHLORINE	KG	29
EPOXIDIZED SOYABEAN OIL	KG	5
CHEM NONYLPHENOL-(RING-UL-14C) 4-ET	KG	0.03
CHEM SODIUM SULPHITE; PN: MIS 19OX	KG	5

HIIR Recipe – B2232/B2247 (Manual Entry)

HIIR Recipe - B2232/B2247	UoM	UOM/MT Normal	UOM/MT Minimum
IRGANOX 1010 / ANOX 20	KG	0.7	0.5
CHEM CALC STEARATE DISPERSION	KG	40	37
CHEM CALCIUM CHLORIDE (CACL2)	KG	2.5	2.5
CHLORINE	KG	20	15.1
EPOXIDIZED SOYABEAN OIL	KG	16	16
CHEM NONYLPHENOL-(RING-UL-14C) 4-ET	KG	0.04	0.04
CHEM SODIUM SULPHITE; PN: MIS 19OX	KG	5.8	5.8
CHEM HYDROCHLORIC ACID HCL 33%	KG	10.4	10.4
SODIUM BROMIDE	KG	45	43.78

For Grade C1139, Cat-Chem Steady State Norms Calculation:

Grade wise steady state norms are derived for following catalysts & chemicals using the logic given by RIL. The detailed logic for norms derivation is captured below for each Cat-Chem particular.

S No	SAP MAT Code	Cat-Chem	S No	SAP MAT Code	Cat-Chem
1	1000025932	HEXANE, POLYMER GRADE	11	1000019006	CHEM FECL3
2	1000000046	IRGANOX 1010 / ANOX 20	12	1000008303	CHEM MAXFLOC-A107
3	1000026599	CHEM CALC STEARATE DISPERSION	13	1000004823	CHEM MAXFLOC C11
4	1000026802	CHEM CALCIUM CHLORIDE (CACL2)	14	1000009916	DIAMMONIUM HYDROGEN PHOSPHATE PURIFIED
5	1000000096	CHEM CHLORINE	15	1000012562	JAGGERY (EDIBLE)
6	1000022731	EPOXIDIZED SOYABEAN OIL	16	1000001839	UREA, NITROGEN CONTENT 46%

7	1000024435	CHEM NONYLPHENOL- (RING-UL-14C) 4-ET	17	1000023520	VASU 4003
8	1000010654	CHEM SODIUM HYDROXIDE NAOH 48- 50%	18	1000023545	VASUFLOC - 5596
9	1000003776	CHEM SODIUM SULPHITE; PN: MIS 19OX	19	1000028273	VASUFLOC - 6329
10	1000010649	CHEM HYDROCHLORIC ACID HCL 33%	20	1000028274	VASUFLOC - 8120

1. HEXANE, POLYMER GRADE

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (MT/MT) = (total consumption quantity, MT) / (total production quantity / 1000, MT)

2. CHEM SODIUM HYDROXIDE NAOH 48-50%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD

- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

3. CHEM SODIUM SULPHITE; PN: MIS 190X

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

4. CHEM HYDROCHLORIC ACID HCL 33%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD

- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

5. CHEM FECL3

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

6. CHEM MAXFLOC-A107

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247

- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

7. CHEM MAXFLOC C11

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

8. DIAMMONIUM HYDROGEN PHOSPHATE PURIFIED

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD

- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

9. JAGGERY (EDIBLE)

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

10. UREA, NITROGEN CONTENT 46%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities

- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

11. VASU 4003

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

12. VASUFLOC - 5596

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

13. VASUFLOC - 6329

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

14. VASUFLOC - 8120

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

15. IRGANOX 1010 / ANOX 20

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade C1139 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade C1139 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding C1139 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total C1139 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIIR-C1139 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

16. CHEM CALC STEARATE DISPERSION

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade C1139 production and its corresponding process order number in a month from SAP production table

- Perform summation of month wise production quantity of grade C1139 from SAP production table
- Filter monthly chemical consumption quantity from SAP consumption table for corresponding C1139 process order number for a given month as identified from SAP production table
- Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total C1139 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIIR-C1139 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

17. CHEM CALCIUM CHLORIDE (CACL2)

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade C1139 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade C1139 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding C1139 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total C1139 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-) 3% of

recipe value of that chemical from HIR-C1139 recipe table. If there are no records found within the range, HIR recipe value shall be the proposed chemical consumption norm for steady state operation

- Take normal arithmetic average of filtered monthly norms from above step

18. CHEM CHLORINE

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Identify grade C1139 production and its corresponding process order number in a month from SAP production table
- Perform summation of month wise production quantity of grade C1139 from SAP production table
- Filter monthly chemical consumption quantity from SAP consumption table for corresponding C1139 process order number for a given month as identified from SAP production table
- Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total C1139 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIR-C1139 recipe table. If there are no records found within the range, HIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

19. EPOXIDIZED SOYABEAN OIL

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.

2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
 - Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade C1139 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade C1139 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding C1139 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total C1139 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIR-C1139 recipe table. If there are no records found within the range, HIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

20. CHEM NONYLPHENOL- (RING-UL-14C) 4-ET

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
 - Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade C1139 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade C1139 from SAP production table

- Filter monthly chemical consumption quantity from SAP consumption table for corresponding C1139 process order number for a given month as identified from SAP production table
- Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total C1139 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIIR-C1139 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

For Grade B2232, Cat-Chem Steady State Norms Calculation:

Grade wise steady state norms are derived for following catalysts & chemicals using the logic given by RIL. The detailed logic for norms derivation is captured below for each Cat-Chem particular.

S No	SAP MAT Code	Cat-Chem	S No	SAP MAT Code	Cat-Chem
1	1000025932	HEXANE, POLYMER GRADE	11	1000019006	CHEM FECL3
2	1000000046	IRGANOX 1010 / ANOX 20	12	1000008303	CHEM MAXFLOC-A107
3	1000026599	CHEM CALC STEARATE DISPERSION	13	1000004823	CHEM MAXFLOC C11
4	1000026802	CHEM CALCIUM CHLORIDE (CACL2)	14	1000009916	DIAMMONIUM HYDROGEN PHOSPHATE PURIFIED
5	1000000096	CHEM CHLORINE	15	1000012562	JAGGERY (EDIBLE)
6	1000022731	EPOXIDIZED SOYABEAN OIL	16	1000001839	UREA, NITROGEN CONTENT 46%
7	1000024435	CHEM NONYLPHENOL-(RING-UL-14C) 4-ET	17	1000023520	VASU 4003
8	1000010654	CHEM SODIUM HYDROXIDE NAOH 48-	18	1000023545	VASUFLOC - 5596

		50%			
9	1000003776	CHEM SODIUM SULPHITE; PN: MIS 19OX	19	1000028273	VASUFLOC - 6329
10	1000010649	CHEM HYDROCHLORIC ACID HCL 33%	20	1000028274	VASUFLOC - 8120
			21	1000026524	CHEM SODIUM BROMIDE, 99.5%

1. HEXANE, POLYMER GRADE

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (MT/MT) = (total consumption quantity, MT) / (total production quantity / 1000, MT)

2. CHEM SODIUM HYDROXIDE NAOH 48-50%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD

- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

3. CHEM SODIUM SULPHITE; PN: MIS 190X

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

4. CHEM HYDROCHLORIC ACID HCL 33%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247

- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

5. CHEM FECL3

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

6. CHEM MAXFLOC-A107

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD

- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

7. CHEM MAXFLOC C11

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

8. DIAMMONIUM HYDROGEN PHOSPHATE PURIFIED

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities

- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

9. JAGGERY (EDIBLE)

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

10. UREA, NITROGEN CONTENT 46%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities

- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

11. VASU 4003

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

12. VASUFLOC - 5596

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

13. VASUFLOC - 6329

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

14. VASUFLOC - 8120

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

15. IRGANOX 1010 / ANOX 20

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
 - Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2232 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2232 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2232 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2232 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIR-B2232 recipe table. If there are no records found within the range, HIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

16. CHEM CALC STEARATE DISPERSION

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
 - Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)

- Identify grade B2232 production and its corresponding process order number in a month from SAP production table
- Perform summation of month wise production quantity of grade B2232 from SAP production table
- Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2232 process order number for a given month as identified from SAP production table
- Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2232 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIR-B2232 recipe table. If there are no records found within the range, HIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

17. CHEM CALCIUM CHLORIDE (CaCl₂)

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2232 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2232 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2232 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step

- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2232 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIIR- B2232 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

18. CHEM CHLORINE

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2232 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2232 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2232 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2232 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIIR- B2232 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

19. EPOXIDIZED SOYABEAN OIL

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2232 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2232 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2232 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2232 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIIR- B2232 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

20. CHEM NONYLPHENOL- (RING-UL-14C) 4-ET

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)

- Identify grade B2232 production and its corresponding process order number in a month from SAP production table
- Perform summation of month wise production quantity of grade B2232 from SAP production table
- Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2232 process order number for a given month as identified from SAP production table
- Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2232 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIR- B2232 recipe table. If there are no records found within the range, HIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

21. CHEM SODIUM BROMIDE, 99.5%

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2232 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2232 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2232 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step

- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2232 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-) 3% of recipe value of that chemical from HIIR- B2232 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

For Grade B2247, Cat-Chem Steady State Norms Calculation:

Grade wise steady state norms are derived for following catalysts & chemicals using the logic given by RIL. The detailed logic for norms derivation is captured below for each Cat-Chem particular.

S No	SAP MAT Code	Cat-Chem	S No	SAP MAT Code	Cat-Chem
1	1000025932	HEXANE, POLYMER GRADE	11	1000019006	CHEM FECL3
2	1000000046	IRGANOX 1010 / ANOX 20	12	1000008303	CHEM MAXFLOC-A107
3	1000026599	CHEM CALC STEARATE DISPERSION	13	1000004823	CHEM MAXFLOC C11
4	1000026802	CHEM CALCIUM CHLORIDE (CACL2)	14	1000009916	DIAMMONIUM HYDROGEN PHOSPHATE PURIFIED
5	1000000096	CHEM CHLORINE	15	1000012562	JAGGERY (EDIBLE)
6	1000022731	EPOXIDIZED SOYABEAN OIL	16	1000001839	UREA, NITROGEN CONTENT 46%
7	1000024435	CHEM NONYLPHENOL-(RING-UL-14C) 4-ET	17	1000023520	VASU 4003
8	1000010654	CHEM SODIUM HYDROXIDE NAOH 48-50%	18	1000023545	VASUFLOC - 5596
9	1000003776	CHEM SODIUM SULPHITE; PN: MIS 19OX	19	1000028273	VASUFLOC - 6329

10	1000010649	CHEM HYDROCHLORIC ACID HCL 33%	20	1000028274	VASUFLOC - 8120
			21	1000026524	CHEM SODIUM BROMIDE, 99.5%

1. HEXANE, POLYMER GRADE

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (MT/MT) = (total consumption quantity, MT) / (total production quantity / 1000, MT)

2. CHEM SODIUM HYDROXIDE NAOH 48-50%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD

- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

3. CHEM SODIUM SULPHITE; PN: MIS 190X

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

4. CHEM HYDROCHLORIC ACID HCL 33%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities

- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

5. CHEM FECL3

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

6. CHEM MAXFLOC-A107

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

7. CHEM MAXFLOC C11

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

8. DIAMMONIUM HYDROGEN PHOSPHATE PURIFIED

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

9. JAGGERY (EDIBLE)

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

10. UREA, NITROGEN CONTENT 46%

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

11. VASU 4003

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

12. VASUFLOC - 5596

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

13. VASUFLOC - 6329

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

14. VASUFLOC - 8120

- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
- Filter C1139 production days where day-wise C1139 production quantity is more than 5 TPD
- Filter B2232 production days where day-wise B2232 production quantity is more than 5 TPD
- Filter B2247 production days where day-wise B2247 production quantity is more than 5 TPD
- Perform summation of day-wise filtered production quantities of C1139, B2232, & B2247
- Filter consumption quantities for the days where day-wise C1139 production value, B2232 production value, & B2247 production value is more than 5 TPD
- Perform summation of day-wise filtered consumption quantities
- Proposed consumption norms (KG/MT) = (total consumption quantity, KG) / (total production quantity / 1000, MT)

15. IRGANOX 1010 / ANOX 20

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2247 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2247 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2247 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2247 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-)3% of recipe value of that chemical from HIIR- B2247 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

16. CHEM CALC STEARATE DISPERSION

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)

- Identify grade B2247 production and its corresponding process order number in a month from SAP production table
- Perform summation of month wise production quantity of grade B2247 from SAP production table
- Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2247 process order number for a given month as identified from SAP production table
- Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2247 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-)3% of recipe value of that chemical from HIIR- B2247 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

17. CHEM CALCIUM CHLORIDE (CACL2)

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2247 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2247 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2247 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step

- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2247 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-)3% of recipe value of that chemical from HIIR- B2247 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

18. CHEM CHLORINE

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2247 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2247 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2247 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2247 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-)3% of recipe value of that chemical from HIIR- B2247 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

19. EPOXIDIZED SOYABEAN OIL

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2247 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2247 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2247 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
 - Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2247 production quantity / 1000, MT)
 - Filter monthly chemical consumption norms which are between the range of (+/-)3% of recipe value of that chemical from HIIR- B2247 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
 - Take normal arithmetic average of filtered monthly norms from above step

20. CHEM NONYLPHENOL- (RING-UL-14C) 4-ET

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)

- Identify grade B2247 production and its corresponding process order number in a month from SAP production table
- Perform summation of month wise production quantity of grade B2247 from SAP production table
- Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2247 process order number for a given month as identified from SAP production table
- Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step
- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2247 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-)3% of recipe value of that chemical from HIIR- B2247 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

21. CHEM SODIUM BROMIDE, 99.5%

1. Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material.
 2. For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data for the duration defined as AOP basis period (1-Apr-2024 till date)
 - Identify grade B2247 production and its corresponding process order number in a month from SAP production table
 - Perform summation of month wise production quantity of grade B2247 from SAP production table
 - Filter monthly chemical consumption quantity from SAP consumption table for corresponding B2247 process order number for a given month as identified from SAP production table
 - Perform summation of month wise chemical consumption quantity of filtered chemical consumption data from above step

- Month wise chemical consumption norm = (month wise total chemical consumption quantity, KG) / (Month wise total B2247 production quantity / 1000, MT)
- Filter monthly chemical consumption norms which are between the range of (+/-)3% of recipe value of that chemical from HIIR- B2247 recipe table. If there are no records found within the range, HIIR recipe value shall be the proposed chemical consumption norm for steady state operation
- Take normal arithmetic average of filtered monthly norms from above step

Utilities (Source System: PAS & SAP):

1. For utilities, steady state consumption quantity is calculated rather than steady state consumption norms using historical data sets from PAS system.
2. Steady state consumption quantity of each utility is calculated separately for each grade i.e. C1139, B2232, & B2247

For grades C1139 & B2232, PAS is the source system to derive the utility norms

Utilities	PAS Tags
Compressed Air	HIIR.UTC002 (Plant Air) + HIIR.UTC003 (Plant Air) + HIIR.UTC004 (Instrument Air) + HIIR.UTC005 (Instrument Air) + HIIR.UTC006 (Breathing Air)
Nitrogen - Other	HIIR.UTC007 (Nitrogen) + HIIR.UTC008 (Nitrogen)
De Mineralized Water	HIIR.UTC016 (DM Water) + HIIR.UTC017 (DM Water)
Power_Dis	HIIR.UTC001
Shp Steam_Dis	HIIR.UTC011

For grade B2247, SAP is the source system to derive the utility norms

Utilities	SAP MAT Code
Compressed Air	COMPARE
Nitrogen - Other	NITROGENO
De Mineralized Water	DM WATER
Power_Dis	310027910
Shp Steam_Dis	310027924

For Grade C1139, Utilities Steady State Consumption Quantity Calculation:

(Source System: PAS)

1. Compressed Air:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where C1139 day-wise production (SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+IIR_OUT010)/1000) is between 160 -180 TPD from last 2 years of PAS historical data
- Filter out compressed air consumption quantities (day wise summation HIIR.UTC002 + HIIR.UTC003 + HIIR.UTC004 + HIIR.UTC005 + HIIR.UTC006) for filtered C1139 production days (minimum 30 data points)
- Filter out production days from PAS where C1139 day-wise production (SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+IIR_OUT010)/1000) is between 100-120 TPD from last 2 years of PAS historical data
- Filter out compressed air consumption quantities (day wise summation HIIR.UTC002 + HIIR.UTC003 + HIIR.UTC004 + HIIR.UTC005 + HIIR.UTC006) for filtered C1139 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered consumption data points to derive the proposed compressed air consumption quantity for steady state operation using following formula:

Compressed Air Proposed Consumption Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise C1139 production quantity})}{\text{SUM}(\text{day wise C1139 production quantity})}$$

2. Nitrogen:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where C1139 day-wise production (SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+IIR_OUT010)/1000) is between 160 -180 TPD from last 2 years of PAS historical data
- Filter out nitrogen consumption quantities (day wise summation HIIR.UTC007 + HIIR.UTC008) for filtered C1139 production days (minimum 30 data points)
- Filter out production days from PAS where C1139 day-wise production (SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+IIR_OUT010)/1000) is between 100-120 TPD from last 2 years of PAS historical data
- Filter out nitrogen consumption quantities (day wise summation HIIR.UTC007 + HIIR.UTC008) for filtered C1139 production days (minimum 30 data points)

- Calculate weighted average of both types of filtered consumption data points to derive the proposed nitrogen consumption quantity for steady state operation using following formula:

Nitrogen Proposed Consumption Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise C1139 production quantity})}{\text{SUM}(\text{day wise C1139 production quantity})}$$

3. De Mineralized Water:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where C1139 day-wise production (SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+IIR_OUT010)/1000) is between 160 -180 TPD from last 2 years of PAS historical data
- Filter out DM water consumption quantities (day wise summation HIIR.UTC016 + HIIR.UTC017) for filtered C1139 production days (minimum 30 data points)
- Filter out production days from PAS where C1139 day-wise production (SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+IIR_OUT010)/1000) is between 100-120 TPD from last 2 years of PAS historical data
- Filter out DM water consumption quantities (day wise summation HIIR.UTC016 + HIIR.UTC017) for filtered C1139 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered consumption data points to derive the proposed DM water consumption quantity for steady state operation using following formula:

DM Water Proposed Consumption Qty (M3)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise C1139 production quantity})}{\text{SUM}(\text{day wise C1139 production quantity})}$$

4. Power:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Daily power consumption from PAS Tag: HIIR.UTC001 (MWH)
- Add 12MWH of additional HIIR CW power in daily power consumption raw data coming from PAS tag: HIIR.UTC001 [Daily power consumption = HIIR.UTC001 + 12MWH]
- Filter out production days from PAS where C1139 day-wise production (SUM(HIIR_OUT006+HIIR_OUT007+HIIR_OUT008+HIIR_OUT009+IIR_OUT010)/1000) is between 160 -180 TPD from last 2 years of PAS historical data
- Filter out daily power consumption quantity for filtered C1139 production days (minimum 30 data points)
- Filter out production days from PAS where C1139 day-wise production

$(\text{SUM}(\text{HIIR_OUT006}+\text{HIIR_OUT007}+\text{HIIR_OUT008}+\text{HIIR_OUT009}+\text{IIR_OUT010})/1000)$ is between 100-120 TPD from last 2 years of PAS historical data

- Filter out daily power consumption quantity for filtered C1139 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered power consumption data sets to derive the proposed power consumption quantity for steady state operation using following formula:

Proposed Power Consumption Qty (KWH)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise C1139 production quantity})}{\text{SUM}(\text{day wise C1139 production quantity})} * 1000$$

5. SHP Steam:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where C1139 day-wise production $(\text{SUM}(\text{HIIR_OUT006}+\text{HIIR_OUT007}+\text{HIIR_OUT008}+\text{HIIR_OUT009}+\text{IIR_OUT010})/1000)$ is between 160 -180 TPD from last 2 years of PAS historical data
- Filter out HHP steam consumption quantities (HIIR_UTC011) for filtered C1139 production days (minimum 30 data points)
- Filter out production days from PAS where C1139 day-wise production $(\text{SUM}(\text{HIIR_OUT006}+\text{HIIR_OUT007}+\text{HIIR_OUT008}+\text{HIIR_OUT009}+\text{IIR_OUT010})/1000)$ is between 100-120 TPD from last 2 years of PAS historical data
- Filter out HHP steam consumption quantities (HIIR_UTC011) for filtered C1139 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered consumption data sets to derive the proposed SHP steam consumption quantity for steady state operation using following formula:

Proposed SHP Steam Consumption Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise C1139 production quantity})}{\text{SUM}(\text{day wise C1139 production quantity})}$$

For Grade B2232, Utilities Steady State Consumption Quantity Calculation:

(Source System: PAS)

1. Compressed Air:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 145 -175 TPD from last 2 years of PAS historical data
- Filter out compressed air consumption quantities (day wise summation HIIR.UTC002 + HIIR.UTC003 + HIIR.UTC004 + HIIR.UTC005 + HIIR.UTC006) for filtered B2232 production days (minimum 30 data points)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 100-120 TPD from last 2 years of PAS historical data
- Filter out compressed air consumption quantities (day wise summation HIIR.UTC002 + HIIR.UTC003 + HIIR.UTC004 + HIIR.UTC005 + HIIR.UTC006) for filtered B2232 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered consumption data points to derive the proposed compressed air consumption quantity for steady state operation using following formula:

Compressed Air Proposed Consumption Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2232 production quantity})}{\text{SUM}(\text{day wise B2232 production quantity})}$$

2. Nitrogen:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 145 -175 TPD from last 2 years of PAS historical data
- Filter out nitrogen consumption quantities (day wise summation HIIR.UTC007 + HIIR.UTC008) for filtered B2232 production days (minimum 30 data points)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 100-120 TPD from last 2 years of PAS historical data

- Filter out nitrogen consumption quantities (day wise summation HIIR.UTC007 + HIIR.UTC008) for filtered B2232 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered consumption data points to derive the proposed nitrogen consumption quantity for steady state operation using following formula:

Nitrogen Proposed Consumption Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2232 production quantity})}{\text{SUM}(\text{day wise B2232 production quantity})}$$

3. De Mineralized Water:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 145 -175 TPD from last 2 years of PAS historical data
- Filter out DM water consumption quantities (day wise summation HIIR.UTC016 + HIIR.UTC017) for filtered B2232 production days (minimum 30 data points)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 100-120 TPD from last 2 years of PAS historical data
- Filter out DM water consumption quantities (day wise summation HIIR.UTC016 + HIIR.UTC017) for filtered B2232 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered consumption data points to derive the proposed DM water consumption quantity for steady state operation using following formula:

DM Water Proposed Consumption Qty (M3)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2232 production quantity})}{\text{SUM}(\text{day wise B2232 production quantity})}$$

4. Power:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Daily power consumption from PAS Tag: HIIR.UTC001 (MWH)
- Add 12MWH of additional HIIR CW power in daily power consumption raw data coming from PAS tag: HIIR.UTC001 [Daily power consumption = HIIR.UTC001 + 12MWH]
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 145 -175 TPD from last 2 years of PAS historical data
- Filter out daily power consumption quantity for filtered B2232 production days (minimum 30

data points)

- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 100-120 TPD from last 2 years of PAS historical data
- Filter out daily power consumption quantity for filtered B2232 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered power consumption data sets to derive the proposed power consumption quantity for steady state operation using following formula:

Proposed Power Consumption Qty (KWH)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2232 production quantity})}{\text{SUM}(\text{day wise B2232 production quantity})} * 1000$$

5. SHP Steam:

- Take 2 years of PAS data (1-Apr-2024 till date)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 145 -175 TPD from last 2 years of PAS historical data
- Filter out HHP steam consumption quantities (HIIR_UTC011) for filtered B2232 production days (minimum 30 data points)
- Filter out production days from PAS where B2232 day-wise production (SUM(HIIR_OUT001+HIIR_OUT002+HIIR_OUT003+HIIR_OUT004+IIR_OUT005)/1000) is between 100-120 TPD from last 2 years of PAS historical data
- Filter out HHP steam consumption quantities (HIIR_UTC011) for filtered B2232 production days (minimum 30 data points)
- Calculate weighted average of both types of filtered consumption data sets to derive the proposed SHP steam consumption quantity for steady state operation using following formula:

Proposed SHP Steam Consumption Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2232 production quantity})}{\text{SUM}(\text{day wise B2232 production quantity})}$$

For Grade B2247, Utilities Steady State Consumption Quantity Calculation:

(Source System: SAP)

1. Compressed Air:

- Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material
- For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter grade B2247 production and its corresponding process order number & posting date from SAP production table where daily wise B2247 production quantity is more than 80000 KG per day (80TPD)
- For a specific utility, filter out consumption quantity of the same B2247 posting date & process order number from SAP
- Calculate weighted average of filtered consumption quantities to derive the steady state consumption quantity of a particular utility

Compressed Air Proposed Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2247 production quantity})}{\text{SUM}(\text{day wise B2247 production quantity})}$$

2. Nitrogen:

- Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material
- For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter grade B2247 production and its corresponding process order number & posting date from SAP production table where daily wise B2247 production quantity is more than 80000 KG per day (80TPD)
- For a specific utility, filter out consumption quantity of the same B2247 posting date & process order number from SAP

- Calculate weighted average of filtered consumption quantities to derive the steady state consumption quantity of a particular utility

Nitrogen Proposed Qty (MT)

$$\frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2247 production quantity})}{\text{SUM}(\text{day wise B2247 production quantity})}$$

3. De Mineralized Water:

- Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material
- For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter grade B2247 production and its corresponding process order number & posting date from SAP production table where daily wise B2247 production quantity is more than 80000 KG per day (80TPD)
- For a specific utility, filter out consumption quantity of the same B2247 posting date & process order number from SAP
- Calculate weighted average of filtered consumption quantities to derive the steady state consumption quantity of a particular utility

DM Water Proposed Qty (M3)

$$\frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2247 production quantity})}{\text{SUM}(\text{day wise B2247 production quantity})}$$

4. Power:

- Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material
- For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter grade B2247 production and its corresponding process order number & posting date from SAP production table where daily wise B2247 production quantity is more than 80000 KG per day (80TPD)

- For a specific utility, filter out consumption quantity of the same B2247 posting date & process order number from SAP
- Calculate weighted average of filtered consumption quantities to derive the steady state consumption quantity of a particular utility

Power Consumption Proposed Qty (KWH)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2247 production quantity})}{\text{SUM}(\text{day wise B2247 production quantity})}$$

5. SHP Steam:

- Process Order Number is assigned to each grade specific product material in a month. In other months, different process order numbers are assigned to the same grade specific product material
- For a given month, process order number for a grade specific product material is used to identify the grade specific product quantity & grade specific consumption quantity in SAP
- Take 2 years of SAP data (1-Apr-2024 till date)
- Filter grade B2247 production and its corresponding process order number & posting date from SAP production table where daily wise B2247 production quantity is more than 80000 KG per day (80TPD)
- For a specific utility, filter out consumption quantity of the same B2247 posting date & process order number from SAP
- Calculate weighted average of filtered consumption quantities to derive the steady state consumption quantity of a particular utility

SHP Steam Proposed Qty (MT)

$$= \frac{\text{SUMPRODUCT}(\text{day wise consumption quantity \& day wise B2247 production quantity})}{\text{SUM}(\text{day wise B2247 production quantity})}$$

Packaging Material (Source: Manual Entry):

- Grade wise packaging material norms shall be manual entry in AOP steady state consumption screen

SAP MAT Code	Packaging Material	Norms UoM	Norms Calculation
1200024302	BAGS, HDPE WOVEN FOR CIIR 1139 INNER LINI	EA	Manual Entry
1200023323	CAP FOR MB-5 CRATE	EA	Manual Entry
1200025829	CRATE LINER, SIZE:180"X55", BLUE	EA	Manual Entry
1200025828	CRATE LINER, SIZE:200"X42", BLUE	EA	Manual Entry
1200026162	CRATE, PLYWOOD, BALE PACKING	EA	Manual Entry
1200031204	CRATE, PLYWOOD, BALE PACKING	EA	Manual Entry
1200026363	EVA FILM, EMBOSSSED, FOR BALE PACKAGING	KG	Manual Entry
1200024328	FILMS, EVA FOR C1139-HIIR WITH NEEDLING	KG	Manual Entry
1200024269	FILMS, EVA FOR CIIR-1139-HIIR BALE PACKIN	KG	Manual Entry
1200024765	LABEL, CRATE, SIZE:150X100 MM- CIIR	EA	Manual Entry
1200015864	LABLE, BARCODE, SIZE:100X40 MM	EA	Manual Entry
1200021641	RIBBON-A, DOMINO BLACK; SIZE:110 MM	EA	Manual Entry
1200000794	RIBBON-A, THERMAL TRANSFER; SIZE:110 MM	ROL	Manual Entry
1200030288	RIBBON-AR, BLACK; SIZE:55MM, V60	EA	Manual Entry

Quality (Source: Manual Entry):

- Grade wise quality norms shall be manual entry in AOP steady state consumption screen

Quality	Norms UoM	Norms Calculation
Prime grade	%	Manual Entry
NP grade	%	Manual Entry
OG grade	%	Manual Entry
Crumb sweep	%	Manual Entry
Plant sweep	%	Manual Entry

Steady State Consumption (Norm/Quantity) (8 Step)

Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: HIR

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Steady State Consumption (Norm/Quantity) GRADE: C1139 (45 Rows)

Particulars	UOM	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
Raw Material													
IR IMPRAMER R...	MT/MT	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	Add
Cat Chem													
IRGANOX 1010 /...	KG /MT	0.70000	0.70000	0.70000	0.70000	0.70000	0.70000	0.70000	0.70000	0.70000	0.70000	0.70000	Add
CHEM CALC STE...	KG /MT	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	Add
CHEM CALCIUM...	KG /MT	2.47795	2.47795	2.47795	2.47795	2.47795	2.47795	2.47795	2.47795	2.47795	2.47795	2.47795	Add
CHEM CHLORINE...	KG /MT	28.38284	28.38284	28.38284	28.38284	28.38284	28.38284	28.38284	28.38284	28.38284	28.38284	28.38284	Add
EPOXIDIZED SOY...	KG/MT	5.00000	5.00000	5.00000	5.00000	5.00000	5.00000	5.00000	5.00000	5.00000	5.00000	5.00000	Add
CHEM NONYLPH...	KG /MT	0.03000	0.03000	0.03000	0.03000	0.03000	0.03000	0.03000	0.03000	0.03000	0.03000	0.03000	Add
HEXANE_POLYM...	MT /MT	0.01503	0.01503	0.01503	0.01503	0.01503	0.01503	0.01503	0.01503	0.01503	0.01503	0.01503	Add
CHEM SODIUM ...	KG /MT	15.95138	15.95138	15.95138	15.95138	15.95138	15.95138	15.95138	15.95138	15.95138	15.95138	15.95138	Add
CHEM SODIUM ...	KG /MT	3.54878	3.54878	3.54878	3.54878	3.54878	3.54878	3.54878	3.54878	3.54878	3.54878	3.54878	Add
CHEM HYDROC...	KG /MT	0.23864	0.23864	0.23864	0.23864	0.23864	0.23864	0.23864	0.23864	0.23864	0.23864	0.23864	Add
CHEM FECL3	KG /MT	0.28767	0.28767	0.28767	0.28767	0.28767	0.28767	0.28767	0.28767	0.28767	0.28767	0.28767	Add
CHEM MAXFLOC...	KG /MT	0.00868	0.00868	0.00868	0.00868	0.00868	0.00868	0.00868	0.00868	0.00868	0.00868	0.00868	Add

Figure 17 Digitized AOP Screen - Grade wise month wise steady state consumption norms

2.1.2.2 Shutdown Consumption Quantity/Norms

- Grade wise shutdown quantity/norm to be added only during the month or period of shutdown via manual entry in digital AOP shutdown consumption screen for HIIR
- For Packaging Material & Quality, shutdown consumption norms are not required to be entered in AOP shutdown consumption screen
- For Raw Material, shutdown consumption norm is not required to be entered in AOP shutdown consumption screen
- For Utilities, grade wise shutdown consumption quantity shall be entered manually in AOP shutdown consumption screen
- For Cat-Chem, grade wise shutdown consumption norms shall be entered manually in AOP shutdown consumption screen

Shutdown Consumption (Quantity) (New)

Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: HIR

ELASTOMER | JMD | HIR

Shutdown Consumption (Norms/Quantity) GRADE: C1139 (45 Rows)

Particulars	UOM	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
Raw Material													
JAGGERY BEHLE	KG /MT	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
UREA NITROGEN...	KG /MT	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
VASU 4003	KG /MT	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
VASUFLOC - 5596	KG /MT	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
VASUFLOC - 6129	KG /MT	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
VASUFLOC - 8120	KG /MT	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
Utility Consumption													
Compressed Air	MT	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
Nitrogen - Other	MT	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
DM WATER	M3	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
Power_Oils	KWH	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
Stop Steam_Oils	MT	0.00000	10.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	Manual Entry...
Packaging Material													
BAG/HOPP W...	EA	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System
CAP FOR MB-E...	EA	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System
CRATE LINER S...	EA	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System

Figure 18 Digitized AOP Screen - Grade wise month wise shutdown consumption norms

2.1.2.3 Slowdown Consumption Quantity/Norms

- Grade wise slowdown quantity/norm to be added only during the month or period of shutdown via manual entry in digital AOP slowdown consumption screen for HIIR
- For Packaging Material & Quality, slowdown consumption norms are not required to be entered in AOP slowdown consumption screen
- For Raw Material, slowdown consumption norm is not required to be entered in AOP slowdown consumption screen
- For Utilities, grade wise slowdown consumption quantity shall be entered manually in AOP slowdown consumption screen
- For Cat-Chem, grade wise slowdown consumption norms shall be entered manually in AOP slowdown consumption screen

Slowdown Consumption (Norm/Quantity) (10 Step)

Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: HIIR

PRODUCTION & N... SHUTDOWN/TA A... SLOWDOWN ACTI... PRODUCTION TAB... BUSINESS DEMAND NET PRODUCTION... MONTH WISE PRO... STEADY STATE CO... SHUTDOWN CON... SLOWDOWN CON... OVERALL AOP CO...

ELASTOMER | JMD | HIIR

Slowdown Consumption (Norms/Quantity) GRADE: B2247 48 Rows

Export Import Save

Particulars	UCM	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Remark
Raw Material													
IIR IMPRAMER R 1675	MT/MT	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System
Cat Chem													
Utility Consumption													
DM WATER	M3	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	Manual Entry...
Power_Dis	KWH	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	Manual Entry...
Compressed Air	MT	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	Manual Entry...
Shp Steam_Dis	MT	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	Manual Entry...
Nitrogen - Other	MT	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	20.00000	Manual Entry...
Packaging Material													
RIBBON-AR, BLACK...	EA	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System
CRATE LINER, SIZE...	EA	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System
FILMS,EVA FOR CIL...	KG	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System
CRATE, PLYWOOD, ...	EA	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System
CRATE, PLYWOOD, ...	EA	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	System

Figure 19 Digitized AOP Screen - Grade wise month wise slowdown consumption norms

2.1.2.4 Budget Norms/ Consumption AOP

The grade wise & particular wise shutdown consumption & slowdown consumption quantities will be summed with the corresponding grade wise & particular wise steady-state operation norms, to derive the budgeted norms which will be inclusive of shutdown & slowdown conditions.

If shutdown consumption is not applicable for any month, its value will be 0. Hence, the above formula is applicable for all the months.

➤ For each packaging material, grade wise overall consumption AOP norms calculation:

C1139 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2232 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2247 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

➤ For each quality particulars, grade wise overall consumption AOP norms calculation:

C1139 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2232 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2247 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

➤ For raw material, grade wise overall consumption AOP norms calculation:

C1139 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2232 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2247 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

➤ For each cat-chem, grade wise overall consumption AOP norms calculation:

C1139 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2232 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

B2247 Overall Consumption AOP Norms = Steady state consumption norm + Shutdown consumption norms + Slowdown consumption norm

➤ For each utility, grade wise overall consumption AOP norms calculation:

*C1139 Overall Consumption AOP Norms = $\frac{\text{Steady state consumption qty} + \text{Shutdown consumption qty} + \text{slowdown consumption qty}}{(\text{HIIR Average Hourly Production rate, TPH} * 24\text{hrs})}$*

*B2232 Overall Consumption AOP Norms = $\frac{\text{Steady state consumption qty} + \text{Shutdown consumption qty} + \text{slowdown consumption qty}}{(\text{HIIR Average Hourly Production rate, TPH} * 24\text{hrs})}$*

*B2247 Overall Consumption AOP Norms = $\frac{\text{Steady state consumption qty} + \text{Shutdown consumption qty} + \text{slowdown consumption qty}}{(\text{HIIR Average Hourly Production rate, TPH} * 24\text{hrs})}$*

Overall consumption AOP norms for individual particulars are calculated and displayed as final norms only for grade 'HIIR Combined' in Overall AOP consumption screen. Calculation is given as follows:

HIIR Combined Norms

$$= \frac{[(C1139 \text{ overall consumption norm} \times C1139 \text{ annual budget production}) + (B2232 \text{ overall consumption norm} \times B2232 \text{ annual budget production}) + (B2247 \text{ overall consumption norm} \times B2247 \text{ annual budget production})]}{(HIIR \text{ Annual Budget Production})}$$

Overall AOP Consumption (Norm/Quantity) (11 Step) Year: 2026-27 Vertical: ELASTOMER Site: JMD Plant: HIIR

< PRODUCTION & N... SHUTDOWN/TA A... SLOWDOWN ACT... PRODUCTION TAR... BUSINESS DEMAND NET PRODUCTION... MONTH WISE PRO... STEADY STATE CO... SHUTDOWN CON... SLOWDOWN CON... OVERALL AOP C >

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Overall AOP Consumption (Norm/Quantity) GRADE: HIIR COMBINED (46 Rows) Export Calculate Release

Particulars	UOM	Apr-26	May-26	Jun-26	Jul-26	Aug-26	Sep-26	Oct-26	Nov-26	Dec-26	Jan-27	Feb-27	Mar-27	Avg.Norms
▼ Raw Material														
IIR IMPRAMER ...	MT/MT	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
▼ Cat Chem														
IRGANOX 1010 /...	KG /MT	24.70558	18.70558	18.70558	18.70558	18.70558	18.70558	18.70558	18.70558	18.70558	18.70558	18.70558	18.70558	19.20558
CHEM CALC ST...	KG /MT	46.00000	52.00000	46.00000	46.00000	46.00000	46.00000	46.00000	46.00000	46.00000	46.00000	46.00000	46.00000	46.50000
CHEM CALCIU...	KG /MT	20.48454	20.48454	20.48454	26.48454	20.48454	20.48454	20.48454	20.48454	20.48454	20.48454	20.48454	20.48454	20.98454
CHEM CHLORINE	KG /MT	43.00086	43.00086	43.00086	43.00086	49.00086	43.00086	43.00086	43.00086	43.00086	43.00086	43.00086	43.00086	43.50086
EPOXIDIZED SO...	KG/MT	27.40000	27.40000	27.40000	27.40000	27.40000	33.40000	27.40000	27.40000	27.40000	27.40000	27.40000	27.40000	27.90000
CHEM NONYLP...	KG /MT	18.03411	18.03411	18.03411	18.03411	18.03411	18.03411	24.03411	18.03411	18.03411	18.03411	18.03411	18.03411	18.53411
CHEM SODIUM ...	KG /MT	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619	29.75619
HEXANE POLY...	MT /MT	18.01503	18.01503	18.01503	18.01503	18.01503	18.01503	18.01503	24.01503	18.01503	18.01503	18.01503	18.01503	18.51503
CHEM SODIUM ...	KG /MT	33.95138	33.95138	33.95138	33.95138	33.95138	33.95138	33.95138	33.95138	39.95138	33.95138	33.95138	33.95138	34.45138
CHEM SODIUM ...	KG /MT	21.54878	21.54878	21.54878	21.54878	21.54878	21.54878	21.54878	21.54878	27.54878	21.54878	21.54878	21.54878	22.04878

Figure 20 Digitized AOP screen for Month-wise Final Consumption AOP for HIIR Combined Grade

3. AOP Reports

- Annual AOP Cost
- Plant Production Summary (T-14)
- Annual Production Plan (T-15)
- Month wise Production Plan (T-16)
- Specific Consumption Norms (T-17)
- Month wise raw data (T-18)
- Turnaround Report (T-19A)
- Shutdown Report (T-19B)
- Shutdown Break-up Last Four Year (T-19C)
- Norms for Shutdown & Slowdown (T-19D)
- Month wise Operating Hours (T-20)
- Plant Contribution (T-21)
- Plant Contribution Summary (T-22)

4. Key Assumptions and Exemptions

The above scope of work should be read with the following assumptions made

1. Data Acceptance
2. DCT provided by IIR_HIIR Site team
3. Site response on Elastomer IIR_HIIR DCT discussion